motors etc for supplying mechanical energy for motion control.

Drives employing electric motors are called electrical drives.

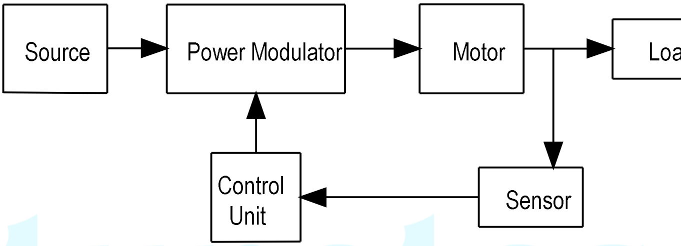


Figure (1)

Source may be a dc or ac (1-phase or 3-phase) source.

Power modulator converts the electrical energy of the source in the motor. For example, if the source is dc and the motor is induction motor, converts dc into a variable frequency ac. Power modulator also limits the cur values during transient operations like starting and braking.

Power modulator is usually a power electric converter like contro inverter, ac voltage controller, cyclo-converter, matrix converter etc. Con gate/base signals for the converters.

PARTS OF ELECTRICAL DRIVES

Major parts are i) Electrical Motor ii) Power Modulator iii) Source iv) Load &

ELECTRICAL MOTORS

i) DC motors – separately excited dc motor, shunt motor, series motor an
ii) Induction Motors – Squirrel Cage Induction Motor, Slip Ring In Induction Motor
iii) Synchronous Motors – Wound field Synchronous Motor (cylindrical r type) & Permanent Magnet Synchronous Motor
iv) Brushless DC Motor
v) Stepper Motor
vi) Switched Reluctance Motor

In the past, induction and synchronous motors were employed mainly for cons Variable speed applications were dominated by dc series motors. Now, ac m variable speed applications because of the development of power electronic cir

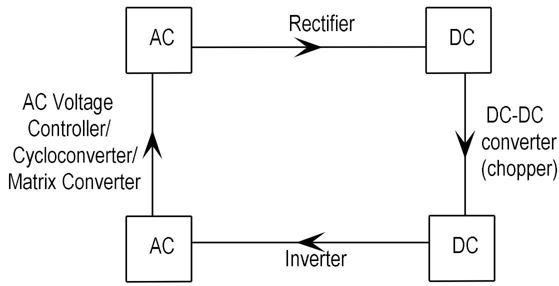
Wound field & permanent magnet synchronous motors have higher full-loa
factor compared to induction motors but cost, size and maintenance is more for
Brushless dc motor is almost similar to permanent magnet synchronous motors
that of conventional brushed dc motor), but has lower cost and requires
converter. It is extensively used in high performance drives, computer dis
robotics, electric vehicles etc.

Stepper motor is popular for position control. A stepper motor rotates by a spec
in response to an input electrical pulse.

Switched reluctance motors have saliency in both stator & rotor. The stator c
with excitation windings on them, and the rotor has salient poles with no w
produced by the tendency of the rotor pole to align with the stator pole to m
linkage when the winding of the stator pole is excited by a current.

Switched reluctance motors (power capability 100W to 100kW) are now com
various applications, ranging from low-power servo motors to high-power tract

POWER MODULATORS



i) AC to DC converters

Single quadrant (motoring) operation is possible with a semi-converter (con
with half number of diodes and half number of thyristors).

Two quadrant (motoring & regenerative braking) operation is possible
(controlled bridge rectifier with thyristors only)

Four quadrant operation is possible with a dual converter.

ii) DC to DC converters (Chopper)

Single quadrant, two quadrant and four quadrant choppers are used.

iii) AC to AC converters

AC voltage controllers can be used for the control of induction motors with s
constant frequency.

is 1-phase 25kV ac, but 1500V dc series motors are usually used. For aircraf
400Hz ac supply is generally used to achieve high power to weight ratio for mo

CONTROL UNIT

Controls for a power modulator are provided by the control unit.

NATURE & CLASSIFICATION OF LOAD TORQUES

i) Load torque may be constant at all speeds,

Example : Cranes during hoisting, compressor

Note: Load torque by compressor that supplies constant pressure system may vary a little with variation of speed.

ii) Load torque proportional to square of the speed

$$T_L = \omega_m^{\ 2}$$

Example : Fans, Centrifugal pumps, propellers

iii) Load torque proportional to speed (this is an uncommon type of load characteristics and is usually observed in a complex form of load)

Example : Separately excited dc generator connected to constant load resistance

For a separately excited dc generator,

$$T_L = \frac{P}{\omega_m} = \frac{VI}{\omega_m} = \frac{k\omega_m \times (k\omega_m / R)}{\omega_m} \propto \omega_m$$

iv) Load torque inversely proportional to the speed

$$T_L \propto \frac{1}{\omega_m} \quad \text{or} \quad T\omega_m = \text{constant (constant power at all speeds)}$$

Example : Paper mill

Note: A fan drive operating at constant speed if, of course, developing constant power. But in applications like paper mill it is desirable to develop constant power over a range of speed.

Compressor, pump and fan type loads required operation in the first qua
operation is unidirectional. They are one quadrant drive systems.

Transportation drives requires operation in both directions. It is a two quadrant

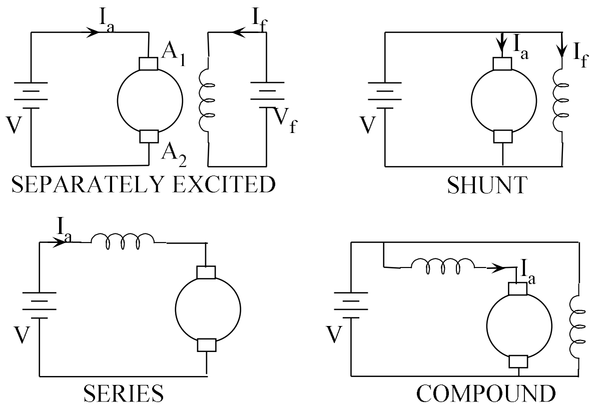
ACTIVE AND PASSIVE TORQUES

Various load torques can be broadly classified into i) active load torque and ii)
Load torques which has the potential to drive the motor under equilibrium con
load torques. Such load torques usually retains their sign when the direction
changed. Example : torque due to gravity. In hoist, when a loaded cage is
torque due to gravity opposes the motion. Therefore, the driving motor has to
overcome the torque due to gravity. On the other hand, when the loaded ca
driven by the torque due to gravity. The motor produces braking torque to lin
safe values.

Load torques which always opposes the motion and change their sign on the
called passive load torques. Example : torque due to friction.

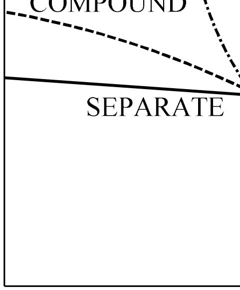
DC MOTOR DRIVES

DC motors can be classified into i) separately excited DC motor ii) shunt motor
iv) compound motor.



In separately excited DC motor, the armature and field are excited by indepen
a dc shunt motor, the field winding is connected in parallel with the armat
supplied from the same source. In series motors, the armature and field are con
supplied from the same source. In compound motor, both series and shunt
performance of the motor is determined by the relative strengths of the series o
In a separately excited motor, it is possible to control both armature voltage an
control the speed over a wide range in the smooth manner. Speeds ranging f

$T = KI_a$



$$\omega_m = \frac{V}{K} - \frac{R_a}{K^2}T$$

For DC series motor,

$$\Phi = K_f I_a$$

$$T = K_e K_f I_a{}^2$$

$$E = K_e K_f I_a \omega_m$$

$$\omega_m = \frac{V}{K_e\Phi} - \frac{I_a R_a}{K_e \Phi} = \frac{V}{\sqrt{K_e K_f}}\frac{1}{\sqrt{T}} - \frac{R_a}{K_e K_f}$$

For series motor, speed is almost inversely proportional to the square root of th
is obtained at low speed and a low torque is obtained at high speed. Series moto
where large starting torques are required, as in cranes, traction, hoists etc.

Problem 1: A 200V, 10.5A, 2000rpm shunt motor has the armature and field re
400Ω respectively. It drives a load whose torque is constant at rated motor torq
speed if the source voltage drops to 175V.

Solution:

$$\Phi_2 = \frac{175}{200}\Phi_1 = 0.875\Phi_1$$

For constant torque, $\Phi_1 I_{a1} = \Phi_2 I_{a2}$

$$I_{a2} = \frac{\Phi_1 I_{a1}}{\Phi_2} = \frac{10.5}{0.875} = 11.4A$$

$$E_1 = V_1 - I_{a1}R_a = 200 - 10.5 \times 0.5 = 195V$$

$$E_2 = V_2 - I_{a2}R_a = 175 - 11.4 \times 0.5 = 169.3V$$

$$\frac{E_1}{E_2} = \frac{\Phi_1 N_1}{\Phi_2 N_2}$$

$$N_2 = \frac{E_2}{E_1}\frac{\Phi_1}{\Phi_2}N_1 = \frac{169.3}{195} \times \frac{1}{0.875} \times 2000 = 1984.5 rpm$$

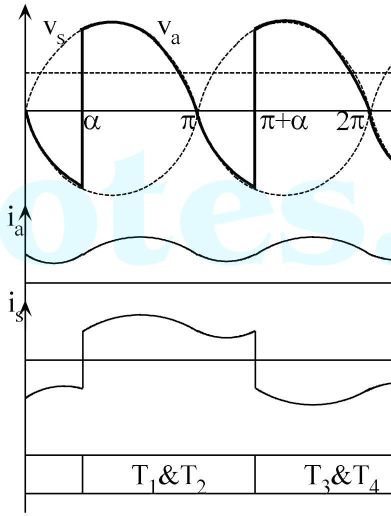
CONTROLLED RECTIFIER FED DC DRIVES

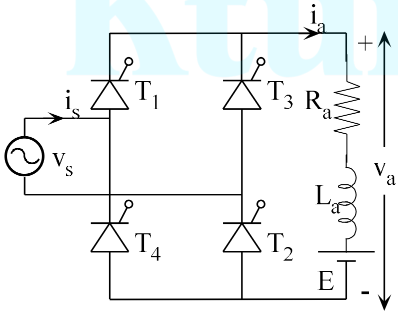
Controlled rectifiers are used to get variable dc voltage from an ac source of fi
rectifiers can be classified into i) 1-phase fully-controlled rectifier ii) 1-phase
iii) 3-phase fully-controlled rectifier and iv) 3-phase half-controlled rec

be continuous.

In continuous conduction mode, armature voltage v_a, armature current i_a an
shown in figure below.

From $\omega t=\alpha$ to $\pi+\alpha$, thyristors T_1 and T_2 are conducting. Armature voltage v_a is same as the source voltage v_s and source current i_s is same as the load current i_a.





At $\omega t=\pi+\alpha$, T_3 and T_4 are turned ON. T_1 and T_2 are turned OFF automatically
applied due to the turning ON of T_3 and T_4. During $\omega t=\pi+\alpha$ to $2\pi+\alpha$, armatu
source current, $i_s = - i_a$.

$$V_a = \int_{\alpha}^{\pi+\alpha} V_m \sin \omega t d(\omega t) = \frac{2V_m}{\pi} \cos \alpha$$

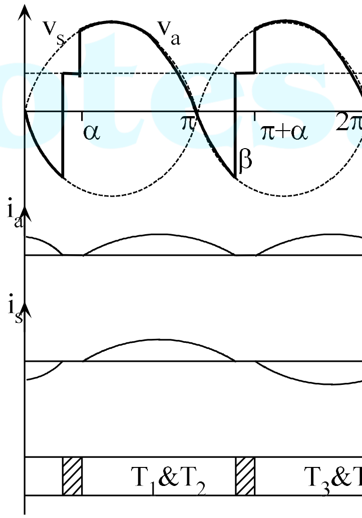
When $\alpha < 90°$, average armature voltage is positive and when $\alpha > 90°$, aver
negative.

In discontinuous conduction mode, armature voltage v_a, armature current i_a ar
shown in figure below.

From $\omega t=\alpha$ to β, thyristors T_1 and T_2 are conducting. Armature voltage v_a is same as the source voltage v_s and source current i_s is same as the load current i_a. At $\omega t=\beta$, armature current i_a decreases to zero and T_1 and T_2 are turned OFF since its anode current falls to zero. Hence, during $\omega t=\beta$ to $\pi+\alpha$, all the devices are OFF. Since armature current is zero, armature voltage is same as the back emf E.

At $\omega t=\pi+\alpha$, T_3 and T_4 are turned ON.

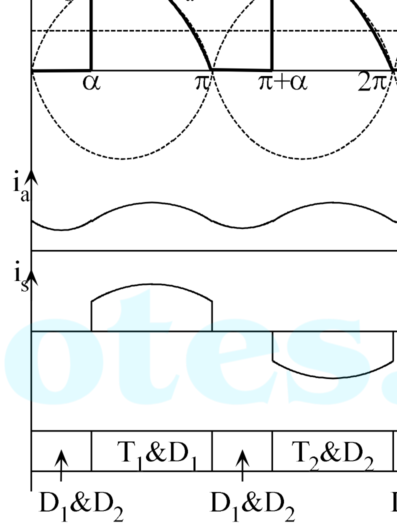
Armature voltage, v_a = -v_s and source current, i_s = - i_a.



Since armature current ia is not perfect dc, the motor torque fluctuates. Sinc
frequency of 100Hz, motor inertia is able to filter out the fluctuations, giving
and ripple-less E.

it starts conducting. At the same instant, T_1 is turned off due to reverse voltage applied by the source through D_2. When D_1 and D_2 are conducting (freewheeling), armature voltage v_a is zero. Source current i_s is also zero.

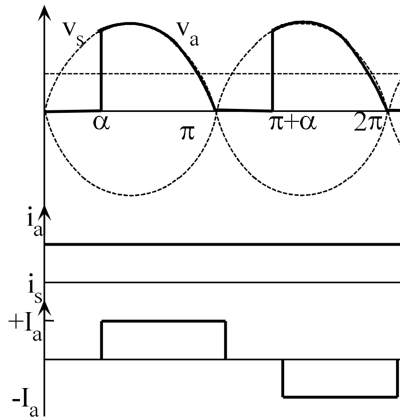
At $\omega t=\pi+\alpha$, T_2 is turned ON by applying a firing pulse. D_1 is automatically turned OFF. During $\omega t=\pi+\alpha$ to 2π, armature voltage, $v_a = -v_s$ and source current, $i_s = -i_a$.



In continuous conduction mode, average armature voltage, V_a is given by

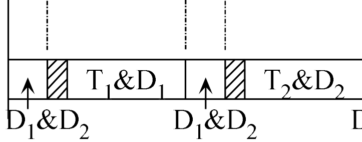
$$V_a = \int_{\alpha}^{\pi} V_m \sin\omega t d(\omega t) = \frac{V_m}{\pi}(1+\cos\alpha)$$

Figure shows the armature voltage v_a, armature current i_a and source current i_s waveforms assuming the armature current is continuous and ripple free ($i_a = I_a$).



In discontinuous conduction mode, armature voltage v_a, armature current i_a ar
shown in figure below.

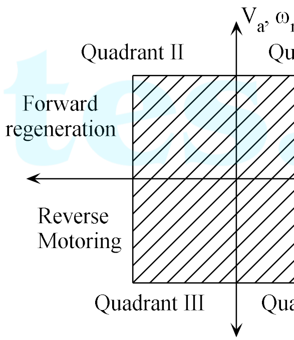
From $\omega t=\alpha$ to π, thyristor T_1 and diode D_2 are conducting. Armature voltage v
voltage v_s and source current i_s is same as the load current i_a.



1-PHASE DUAL CONVERTER (FOUR QUADRANT CONVERTER)

Semiconverter operates in one quadrant (Quadrant I) only and full converter operates in two quadrants (Quadrant I & IV). Dual converters can operate in all the four quadrants (output voltage and current may be either positive or negative).

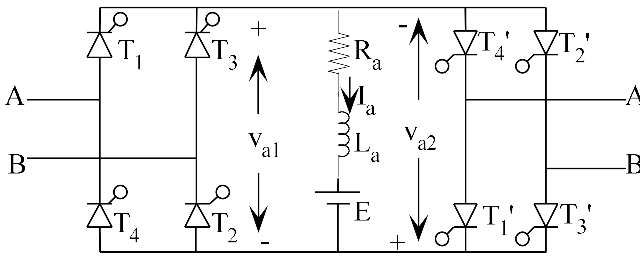
In dual converter, two single phase or three phase fully controlled rectifiers are connected in anti-parallel.



2-types :- i) Nor-circulating current type (non-simultaneous type) and ii) circula
(simultaneous type)

DUAL CONVERTER WITHOUT CIRCULATING CURRENT

Assumption :- Load current is continuous



Only one converter is operated at a time and it alone carries the entire load curr
(Note : Positive current is possible with converter 1 and negative current is pos
Initially let the drive be in operation in quadrant I. Converter 1 is operated as
firing pulses to the converter 2 is blocked. Motor runs in forward direction. Fo
must initially operate in the second quadrant and then in third quadrant. For t

i) Proper protection is necessary for avoiding a short circuit due to r
converters are simultaneously operated.
ii) A delay is required to shift the operation from one mode to the other m

DUAL CONVERTER WITH CIRCULATING CURRENT

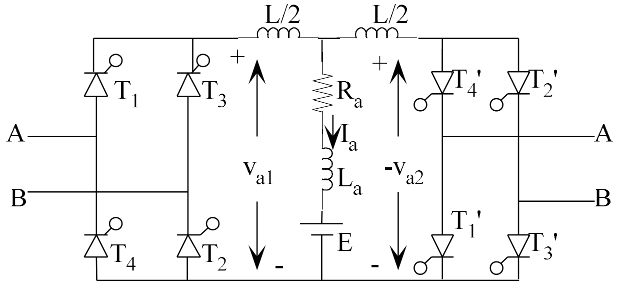
Two converters are operated simultaneously so that the sum of their average ter
so that no dc current circulates in the loop formed by the two converters.

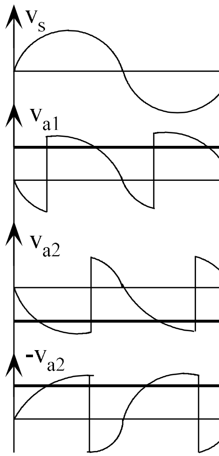
$$V_{a1} + V_{a2} = 0$$

$$\frac{2V_m}{\pi}\cos\alpha_1 + \frac{2V_m}{\pi}\cos\alpha_2 = 0$$

$$\cos\alpha_1 = -\cos\alpha_2 = \cos(180 - \alpha_2)$$

$$\alpha_1 + \alpha_2 = 180°$$





Hence, when one converter operates as rectifier with firing angle α_1, the other is operated as inverter with firing angle $(180-\alpha_1)$. Though their average output voltages are equal, instantaneous voltages v_{a1} and v_{a2} may be different. This results in a voltage difference and hence a large circulating current flows between the two converters but not through the load.

This circulating current is limited to a tolerable value by inserting a rea
converters.

Although both thyristors are operated simultaneously, motor control in first a
provided by converter I (I_a positive & V_{a1} positive or negative). Converter II
current. The circulating current is limited to 15 to 30% of the rated arma
converters reverse their roles when the operating takes place in the 2nd and 3rd q

Advantages : i) No short circuit; ii) better dynamic response

$$\frac{3\sqrt{3}V_m \cos\alpha_1}{\pi} + \frac{3\sqrt{3}V_m \cos\alpha_2}{\pi} = 0$$

$$\cos\alpha_1 = -\cos\alpha_2 = \cos(180 - \alpha_2)$$

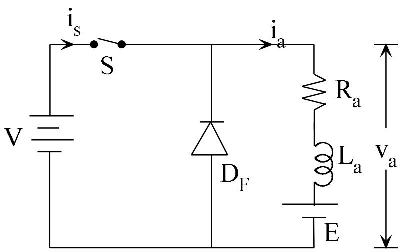
$$\alpha_1 + \alpha_2 = 180°$$

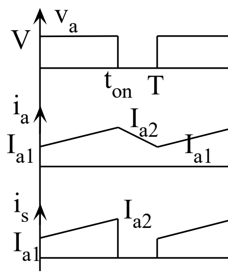
CHOPPER CONTROLLED SEPARATELY EXCITED DC MOTOR

Choppers are used for the control of DC motors because of number of ad efficiency, flexibility in control, light weight, small size, quick response, anc very low speeds. Chopper controlled DC motors have applications in servos anc Chopper offers a number of advantages over controlled rectifiers. Because of t the output voltage ripple, the ripple in the motor armature current is less discontinuous conduction in the speed-torque plane is smaller. Reduction i ripple reduces the machine losses and its derating. A reduction or elimin conduction region improves speed regulation and transient response of a drive.

A chopper can be operated at comparatively high frequencies. With powe frequency can be 2.5kHz and with power MOSFETs, operating frequency car rectifier output voltage and current have a much lower frequency – 100Hz ir rectifier and 300Hz in the case of a 3-phase fully-controlled rectifier-when the 50Hz.

MOTORING CONTROL (CLASS A CHOPPER)

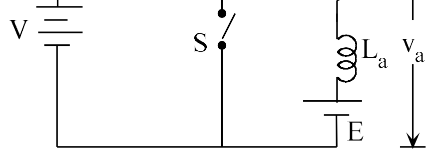


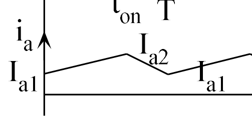


armature voltage, ar
and input cu

During ON period of the switch, armature voltage, $v_a = V$. Armature current increases from I_{a1} to I_{a2}.

$$R_a i_a + L_a \frac{di_a}{dt} + E = V, \quad 0 \le t \le t_{on}$$





armature voltage &

During ON period of the switch, armature voltage, $v_a = 0$. Armature current i
DC machine works as a generator and a part of the energy is stored in the ar
remainder is dissipated in armature resistance.

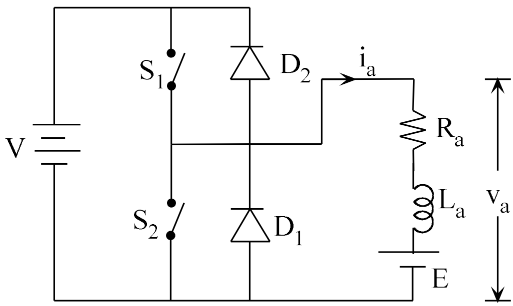
When switch is turned off, armature current flows through diode D and source
decreases from I_{a2} to I_{a1}. The stored energy in the inductance and the energy s
is fed to the source.

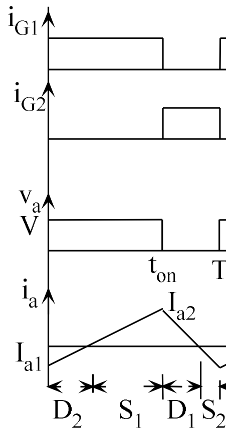
Duty ratio, $D = \dfrac{T - t_{on}}{T}$

Average armature voltage, $V_a = \dfrac{1}{T}\displaystyle\int_{t_{on}}^{T} V dt = DV$

Speed, $\omega_m = \dfrac{DV}{K} - \dfrac{R_a}{K^2}T$

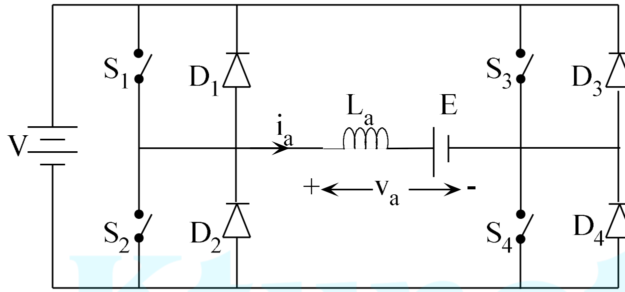
TWO QUADRANT CHOPPER (CLASS C)

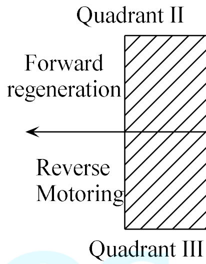




Although switching from class A to class B configuration is a satisfactory
regenerative braking for some applications, in others a smooth transition from

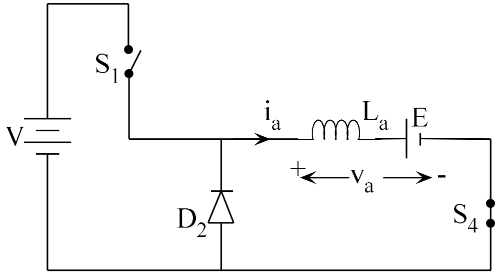
FOUR QUADRANT CHOPPER (CLASS E CHOPPER)





In this chopper, both the armature voltage and armature current may be either p

Equivalent circuits and devices conducting in each quadrant is as shown below.



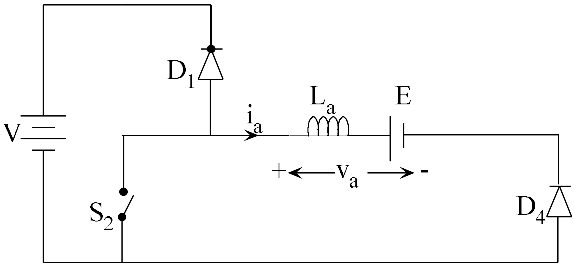
QUADRANT 1

S_2 & S_3 are kep

S_4 is kept ON;

S_1 is operated C

V_a and I_a are po



QUANDRANT

S_1, S_3 & S_4 are

S_2 is operated C

V_a is positive ar

NOTE : For Quadrant 1, S_1 is operated ON & OFF; for Quadrant 2, S_2 is operat
Quadrant 3, S_3 is operated ON & OFF; for Quadrant 4, S_4 is operated ON & OF

AC DRIVES

SPEED CONTROL OF 3-PHASE INDUCTION MOTOR

For an induction motor,

Synchronous speed, $\omega_s = \dfrac{4\pi f}{P}$ rad/sec

Motor speed, $\omega_m = (1-s)\omega_s$

Basic methods of speed control of induction motor are

i) By changing the number of poles
ii) By varying the supply frequency
iii) By varying the stator voltage
iv) By varying the rotor resistance
v) By injecting emf of slip frequency into the rotor circuit

Method (i) is applicable only for squirrel cage induction motors and (iv) an
induction motors.

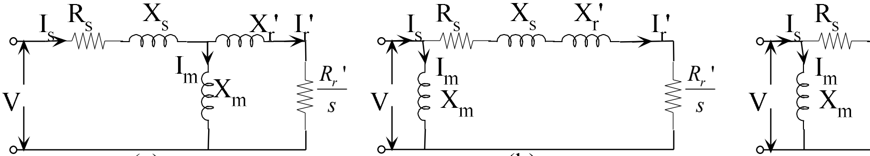
TORQUE EQUATION

Equivalent circuit of the induction motor is shown below. R_r' and X_r' are the ro
rotor reactance X_r referred to stator.

$s = \dfrac{\omega_s - \omega_m}{\omega_s}$ where ω_m = rotor speed in rad/sec and ω_s = synchronous speed in r

$$\omega_s = \frac{4\pi f}{P} \text{ rad/sec}$$

$$\omega_m = (1-s)\omega_s$$



(a) (b)

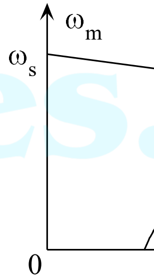
ds

Slip at maximum torque, $s_m = \dfrac{R_r'}{\sqrt{R_s^2 + (X_s + X_r')^2}}$

Maximum (break-down) torque, $T_{\max} = \dfrac{3}{2\omega_s} \dfrac{V^2}{R_s + \sqrt{R_s^2 + (X_s + X_r')^2}}$

Note: Maximum torque is independent of rotor resistance.

T_{max} is independent of rotor resistance but s_m is directly proportional to rotor resistance.



Drop in speed from no-load to full load depends on the rotor resistance. When rotor resistance is low, the drop is quite small, and therefore, motor operates essentially at a constant speed.

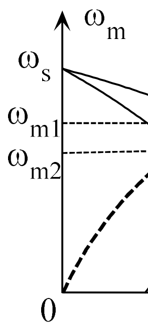
For slips much smaller than s_m, the speed-torque characteristics is a straight line. For slips much larger than s_m, torque is inversely proportional to slip and hence has hyperbolic shape.

STATOR VOLTAGE CONTROL

By reducing stator voltage, speed of a high-slip induction motor can be reduced by an amount which is sufficient for the speed control of fan and pump drives ($T_L = \omega_m^2$). Torque is directly proportional to the square of the applied voltage.

$$T = \frac{3}{\omega_s} \frac{V^2 R_r'/s}{(R_s + R_r'/s)^2 + (X_s + X_r')^2}$$

The speed-torque characteristics for different stator voltages are shown in figure.



The speed can be varied from ω_{m1} to ω_{m2}. The slip s_m for maximum torque i voltage.

Maximum torque, $T_{\max} = \dfrac{3}{2\omega_s} \dfrac{V^2}{(X_s + X_r')}$ if stator resistance is neglected (R_s = 0

If the voltage is reduced to 80%, the maximum torque falls to 64%.

The synchronous speed corresponding to the rated frequency is called bas
control below base speed, while frequency is reduced the voltage is also reduc
V/f ratio is constant. In constant V/f control, flux remains constant at its rated v
frequency, magnetizing current remains constant while core loss decreases. Ma
constant.

The synchronous speed at any other frequency, $\omega_s = k\omega_b$ where $k < 1$ and slip

$$T = \frac{3}{k\omega_b} \frac{(kV)^2 R_r'/s}{(R_s + R_r'/s)^2 + (kX_s + kX_r')^2}$$ where V = rated voltage, X_s and X_r'
reactances at rated frequency

Maximum torque at rated frequency, $T_{\max} = \frac{3}{2\omega_b} \frac{V^2}{R_s + \sqrt{R_s^2 + (X_s + X_r')^2}}$

If R_s is neglected, $T_{\max} = \frac{3}{2\omega_b} \frac{V^2}{X_s + X_r'}$

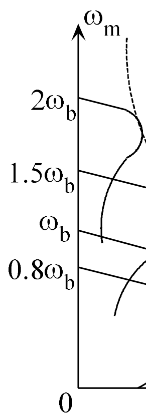
Maximum torque at reduced frequency $T_{\max} = \frac{3}{2k\omega_b} \frac{(kV)^2}{k(X_s + X_r')} = \frac{3}{2\omega_b} \frac{V^2}{(X_s + X_r'}$

Hence, for speed control below based speed, the maximum torque remains con
constant.

For speed control above base speed (k > 1), voltage is kept constant at its rated value while increasing the frequency above rated frequency. Since V/f ratio decreases, flux decreases. The maximum torque is given by,

$$T_{\max} = \frac{3}{2k\omega_b} \frac{V^2}{k(X_s + X_r')} = \frac{3}{2\omega_b (X_s + X_r')} \left(\frac{V}{k}\right)^2$$ where k > 1.

Maximum torque decreases by factor k^2 where k is greater than 1. (Maximum torque is inversely proportional to frequency squared similar to the behavior of dc series motors. In this mode of control, the motor is said to be operated in field-weakening mode.



The voltage at variable frequency can be obtained by a 3-phase cycloconve
source/current source inverter (see 3-phase sine PWM inverter). The cycloco

At 30Hz, $\omega_s = k\omega_b = 0.5 \times 188.5 = 94.25 rad/s$

$$T_{\max} = \frac{3}{2k\omega_b} \frac{(kV)^2}{R_s + \sqrt{R_s^2 + k^2(X_s + X_r')^2}} = \frac{3}{2 \times 0.5 \times 188.5} \frac{(0.5 \times 460/\sqrt{3})^2}{0.66 + \sqrt{0.66^2 + 0.5^2 \times 2.85^2}}$$

$$s_m = \frac{R_r'}{\sqrt{R_s^2 + k^2(X_s + X_r')^2}} = \frac{0.38}{\sqrt{0.66^2 + 0.5^2 \times 2.85^2}} = 0.242$$

$$\omega_m = (1 - s_m)\omega_s = (1 - 0.242) \times 0.5 \times 188.5 = 71.44\, rad/\sec \text{ or } 682\text{rpm}$$

(b) If R_s is neglected,

At 60Hz, $T_{\max} = \frac{3}{2\omega_s} \frac{V^2}{(X_s + X_r')} = \frac{3}{2 \times 188.5} \frac{(460/\sqrt{3})^2}{2.85} = 196.94 Nm$

$$s_m = \frac{R_r'}{(X_s + X_r')} = \frac{0.38}{2.85} = 0.133$$

At 30Hz, $T_{\max} = \frac{3}{2k\omega_b} \frac{(kV)^2}{k(X_s + X_r')} = \frac{3}{2 \times 0.5 \times 188.5} \frac{(0.5 \times 460/\sqrt{3})^2}{0.5 \times 2.85} = 196.94 Nm$

$$s_m = \frac{R_r'}{0.5(X_s + X_r')} = \frac{0.38}{0.5 \times 2.85} = 0.267$$

$$\omega_m = (1 - s_m)\omega_s = (1 - 0.267) \times 0.5 \times 188.5 = 61.09\, rad/\sec \text{ or } 659.7\text{rpm}$$

FIELD ORIENTED CONTROL (VECTOR CONTR

PRINCIPLE

Advantages of DC drives over AC drives using conventional inverter-fed induc

i) Good transient response of the DC motor
ii) Torque can be directly controlled even under transient conditions by c current.

In a separately excited dc motor, electromagnetic torque is proportional to th and armature current. If magnetic saturation is neglected, field flux is proportio is unaffected by the armature current because of orthogonal (field flux and apart) displacement. Hence for a separately excited dc motor, if the field curre torque will be proportional to the armature current. The field flux and the de independently controlled.

PART A (3 x 10 = 30 Marks)

Answer all Questions. Each question carries 3 Mar

1. Explain different turn on methods of SCR.

2. Describe the reverse recovery characteristics of a power diode.

3. Draw the input and output voltage waveforms of single phase half cont
RL load in continuous and discontinuous conduction mode.

4. Explain with neat sketches, the input and output voltage waveforms of
rectifier with R load for a firing angle of 30^0.

5. Compare voltage source and current source inverters.

6. Explain the terms modulation index and frequency modulation ratio rel
modulation.

7. Explain time ratio control method to vary the output voltage in choppe

8. Derive the expression for output voltage of a Buck Converter.

9. What are the advantages of electric drives?

10. Explain regenerative braking control in drives.

PART B (14 x 5 = 70 Marks)

Answer any one full question from each module. Each questi

Module 1

11. a) Explain the two transistor analogy of SCR.

    b) Compare the switching characteristics of IGBT.

12. a) Explain the structural details of MOSFET.

    b) Write short note on wideband gap devices.

b) Write short notes of THD.

16. a) Explain sinusoidal PWM technique for varying the magnitude of out
phase inverter.

b) Briefly explain current source inverter

Module 4

17. a) Explain the working of a Buck-Boost regulator, showing relevant wa
the expression for its output voltage.

b) Design a DC-DC Converter with 12 V input and 200 V output at upt
the output voltage and input current should not exceed +- 5% and +-
Select suitable device and switching frequency.

18. a) Describe the working of four quadrant chopper in all the four quadr
circuit diagrams.

b) Briefly explain the current limit control in dc-dc converter

Module 5

19. a) Explain the working of a single-phase full converter drive

b) Explain the working of a four-quadrant chopper drive

20. a) Explain the stator voltage control for Induction motor drive

b) Explain the working of v/f control of Induction motor drive